Data Alert

The CHARLS Team

February 4, 2023

The CHARLS 2018 wave included a module, CHARLS-HCAP, intended to assess the cognitive status of our respondents 60 years and older. It has come to our attention that one of our assessment instruments, the Mini-Mental State Examination (MMSE), which is a widely used cognition assessment, is likely biased because the Chinese version of MMSE was validated only among people who can read and write and those without vision impairments and in mandarin.

In particular, the following questions require literacy:

- `dc019_w4`: Read the words on the screen, then do what it says.
- `dc023_w4`: Write any complete sentence
- `dc024_w4`: Copy the overlapping pentagons

The following questions require vision:

- `dc019_w4`: Read the words on the screen, then do what it says.
- `dc016_w4`: Naming a watch
- `dc017_w4`: Naming a pencil
- `dc024_w4`: Copy the overlapping pentagons

In addition, word lists used for memory tests are paraphrased in mandarin and may disadvantage non-mandarin speakers:

- `dc013_w4_1` and `dc015_w4`: Word list recalls
- `dc018_w4`: Repeating a phrase

Additionally, some other items in the CHARLS-HCAP may be biased as well. One example is an HRS-TICS item, "`dc026_w4`. What do you call the kind of prickly plant that grows in the desert?" Because the cactus is a less-known plant in China, many people failed to answer correctly, especially less educated people. Another example is CERAD 10-word recall. Some words on the list, such as butter and queen, are not familiar to many Chinese, especially among less educated people.

We are working on detecting biases of MMSE and other instruments against the illiterate, vision impaired and non-mandarin speakers and are finding ways to correct the biases. Once this is finished, we will update the data. Meanwhile, users are advised to use the data with caution.